An Alternative Semianalytical/Analytical Solution to Field-to-Wire Coupling in an Electrically Large Cavity

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***Abstract*—In this paper, we propose an alternative semianalytical solution to problems of field-to-wire coupling in an electrically large cavity employing a different method of derivation. This model is based on transmission-line (TL) theory and the plane wave integral representation for the fields as well as the statistical properties of the fields in a well-stirred reverberation chamber. Compared to an earlier semianalytical solution which is a sum of double integrals over the length and/or the height of the TL, this solution is also a sum of double integrals but over the solid angle. In the earlier one, some terms were neglected due to a simplification, whereas the solution proposed here is a complete one without neglecting any terms. However, when the neglected terms are included in the earlier one, these two solutions are equivalent to each other. Furthermore, directly from this semianalytical solution, we also derive a closed-form solution under some reasonable approximations. Although the solutions are derived for the single-wire configuration, they can be converted to that for the double-wire configuration as well. The results obtained by the aforementioned methods and the other methods are compared with each other in several numerical examples.**

***Index Terms*—Analytical solution, electromagnetic (EM) coupling, field-to-wire, reverberation chamber, semianalytical solution, stochastic electromagnetic (EM) fields, transmission lines (TLs).**

I. Introduction

SINCE cables and wires are often the most susceptible parts in electronic and electric equipments, there have been a lot of research works focusing on field-to-wire coupling in electromagnetic (EM) compatibility field. Typically, in [1], a simple model of field-to-wire coupling was included and widely referenced.

Due to the fact that the frequency of interest is higher and higher, the wavelength of the incident EM wave is comparable to or much smaller than the dimensions of the device. To study problems in these electrically large systems like cavities, a statistical method was well established and applied [2]–[4]. Furthermore, based on the statistical properties of electrically large cavities and transmission-line (TL) theory [5], some simple problems of field-to-wire coupling in an electrically large cavity were solved, as presented in [6] and [7].

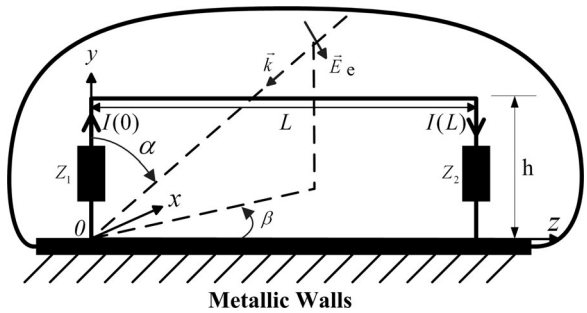


Fig. 1. Model of field-to-wire coupling in a reverberation chamber.

Different from [6], in this paper, we take another method to get an alternative form of semianalytical solution to the same model as in [6] without involving any spatial correlation function. Furthermore, under some reasonable approximations, a closed-form solution is derived directly from the semianalytical one proposed in this paper. We will see that the analytical form is more general than that in [7] and [8] in which the analytical solution was considered to be available only in matched cases. From the analytical solution, we can understand the mechanism of field coupling to wire much more easily and clearly.

This paper is organized as follows. Section II covers the often used EM coupling model based on the TL theory. Section III presents the derivation of the semianalytical solution. Section IV briefly discusses the relationship between the solution proposed here and the earlier one in [6] and gives the analytical solution after a simple derivation. In Section V, some simulated results based on different methods are compared with each other.

II. EM Coupling Model Based on TL Theory

As shown in Fig. 1, a single perfectly electrically conducting (PEC) thin wire is running along a wall of a reverberation chamber in which the walls are considered to be PEC. This wire is situated in the $x = 0$ plane and is parallel to the z-axis. It is loaded at both ends by impedances Z_1 and Z_2. The wire length is L, its radius is r, and its height above the closest wall is h.

When this wire is illuminated by an EM field, the induced currents at both ends are given by Baum–Liu–Tesche equation

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[1] and can be written as follows:

$$I(0)=\frac{1}{Z_c}\frac{1-\rho_1}{\mathrm{e}^{\gamma L}-\rho_1\rho_2\mathrm{e}^{-\gamma L}}\left(\mathrm{e}^{\gamma L}Q-\rho_2\mathrm{e}^{-\gamma L}P\right) \quad (1)$$

$$I(L)=\frac{1}{Z_c}\frac{1-\rho_2}{\mathrm{e}^{\gamma L}-\rho_1\rho_2\mathrm{e}^{-\gamma L}}\left(\rho_1 Q-P\right) \quad (2)$$

where

$$\begin{aligned}
&Q=Q_y^1+Q_z+Q_y^2,\quad P=P_y^1+P_z+P_y^2\\
&Q_y^1=-\frac{1}{2}\int_0^h E_y(0,y,0)dy,\quad P_y^1=-\frac{1}{2}\int_0^h E_y(0,y,0)dy\\
&Q_z=-\frac{1}{2}\int_0^L \mathrm{e}^{-\gamma z}E_z(0,h,z)dz,\\
&P_z=-\frac{1}{2}\int_0^L \mathrm{e}^{\gamma z}E_z(0,h,z)dz\\
&Q_y^2=\frac{1}{2}\mathrm{e}^{-\gamma L}\int_0^h E_y(0,y,L)dy,\\
&P_y^2=\frac{1}{2}\mathrm{e}^{\gamma L}\int_0^h E_y(0,y,L)dy\\
&\rho_1=(\mathrm{Z}_1-Z_\mathrm{c})/(\mathrm{Z}_1+Z_\mathrm{c}),\quad \rho_2=(\mathrm{Z}_2-Z_\mathrm{c})/(\mathrm{Z}_2+Z_\mathrm{c})
\end{aligned} \quad (3)$$

where Z_c is the characteristic impedance of the TL, γ is the propagation constant along the TL, both $E_z(\bullet)$ and $E_\mathrm{y}(\bullet)$ given in [6, eq. (5)] are the total tangential electric fields along the wire parallel to the z-axis and the y-axis, respectively.

It has to be noted that in order to obtain precise currents predicted by (1) and (2) at terminals, the TL cross section has to be electrically small at the frequency of interest [5].

III. Proposed Solution

In this section, the plane wave integral representation for the fields and some statistical properties of the fields in reverberation chambers are applied to the model shown in Fig. 1, leading to the mean square of the magnitude of the induced currents at both ends semianalytically.

The plane wave integral representation originally from [9] for the fields in reverberation chambers was well established by Hill [3]. Based on the plane wave integral representation and some appropriate statistical assumptions, some interesting properties of the fields as well as the other quantities were derived. When these statistical properties were applied to the field-to-wire coupling equations, as in [6], a semianalytical solution to problems of field-to-wire coupling was obtained. Slightly different from [3] but without loss of generality, we use the coordinate system as shown in Fig. 1. Therefore, the vector wavenumber $\vec{k}=-k(\hat{x}\sin\alpha\sin\beta+\hat{y}\cos\alpha+\hat{z}\sin\alpha\cos\beta)$, where $k=\omega\sqrt{\mu\varepsilon}$ is the free-space wavenumber, μ is the free-space permeability, and ε is the free-space permittivity. Details about the angular spectrum $F(\Omega)$ and the chamber constant E_0^2 are in [3] and [4].

As can be seen from [3, eq. (1)], the total electric field in reverberation chambers can be a superposition of plane waves from all real angles. In other words, the TL near a wall of a reverberation chamber is directly exposed to a continuous distribution of uncorrelated plane waves of infinitesimal amplitude arriving from every possible direction in the upper half-space. Accordingly, the infinitesimal complex amplitude of the plane wave impinging from direction Ω can be represented as $F(\Omega)d\Omega$, where $d\Omega=\sin\alpha d\alpha d\beta$ is the elemental solid angle. Therefore, the total induced currents $I^t(0)$ and $I^t(L)$ at both ends are the sum of contributions of all plane waves and can be written as integrals

$$I^t(0)=\iint_\Omega dI(0),\ I^t(L)=\iint_\Omega dI(L) \quad (4)$$

where $\Omega=2\pi$, and

$$\begin{aligned}
&dI(0)=\frac{1}{Z_c}\frac{1-\rho_1}{\mathrm{e}^{\gamma L}-\rho_1\rho_2\mathrm{e}^{-\gamma L}}\left(\mathrm{e}^{\gamma L}dQ-\rho_2\mathrm{e}^{-\gamma L}dP\right)\\
&dI(L)=\frac{1}{Z_c}\frac{1-\rho_2}{\mathrm{e}^{\gamma L}-\rho_1\rho_2\mathrm{e}^{-\gamma L}}\left(\rho_1 dQ-dP\right)\\
&dQ=dQ_y^1+dQ_z+dQ_y^2,\quad dP=dP_y^1+dP_z+dP_y^2.
\end{aligned} \quad (5)$$

After a simple analytical operation, we have

$$\begin{aligned}
dP_y^1&=dQ_y^1=F_\alpha\sin\alpha\sin(kh\cos\alpha)/(k\cos\alpha)d\Omega\\
dQ_z&=-i(F_\alpha\cos\alpha\cos\beta-F_\beta\sin\beta)\sin(kh\cos\alpha)\\
&\quad\bullet\left[\mathrm{e}^{-(\gamma+ik\sin\alpha\cos\beta)L}-1\right]/(\gamma+ik\sin\alpha\cos\beta)d\Omega\\
dP_z&=i(F_\alpha\cos\alpha\cos\beta-F_\beta\sin\beta)\sin(kh\cos\alpha)\\
&\quad\bullet\left[\mathrm{e}^{(\gamma-ik\sin\alpha\cos\beta)L}-1\right]/(\gamma-ik\sin\alpha\cos\beta)d\Omega\\
dQ_y^2&=-\mathrm{e}^{-(\gamma+ik\sin\alpha\cos\beta)L}F_\alpha\sin\alpha\sin(kh\cos\alpha)\\
&\quad/(k\cos\alpha)d\Omega\\
dP_y^2&=-\mathrm{e}^{-(-\gamma+ik\sin\alpha\cos\beta)L}F_\alpha\sin\alpha\sin(kh\cos\alpha)\\
&\quad/(k\cos\alpha)d\Omega.
\end{aligned} \quad (6)$$

where i is the imaginary unit.

In most cases, the mean of the squared magnitude of the currents at the ends is of interest and can be expressed as follows:

$$\begin{aligned}
\left\langle\left|I^t(0)\right|^2\right\rangle&=\left\langle I^t(0)\bullet I^{t*}(0)\right\rangle\\
&=\left|\frac{1}{Z_c}\frac{1-\rho_1}{\mathrm{e}^{\gamma L}-\rho_1\rho_2\mathrm{e}^{-\gamma L}}\right|^2\\
&\quad\times\left\langle\left(\left|Q^t\right|^2+\left|\rho_2\right|^2\left|P^t\right|^2-2\mathrm{Re}(\rho_2\mathrm{e}^{-2\gamma L}P^tQ^{t*})\right)\right\rangle
\end{aligned} \quad (7)$$

$$\begin{aligned}
\left\langle\left|I^t(L)\right|^2\right\rangle&=\left\langle I^t(L)\bullet I^{t*}(L)\right\rangle\\
&=\left|\frac{1}{Z_c}\frac{1-\rho_2}{\mathrm{e}^{\gamma L}-\rho_1\rho_2\mathrm{e}^{-\gamma L}}\right|^2\\
&\quad\times\left\langle\left(\left|\rho_1\right|^2\left|Q^t\right|^2+\left|P^t\right|^2-2\mathrm{Re}(\rho_1^*P^tQ^{t*})\right)\right\rangle
\end{aligned} \quad (8)$$

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where $\langle\bullet\rangle$ is the expectation operator meaning the mean value of the variable, $*$ denotes complex conjugate, and

$$Q^t = \iint_\Omega dQ = \iint_\Omega (dQ_y^1 + dQ_z + dQ_y^2) = Q_y^{t1} + Q_z^t + Q_y^{t2} \tag{9}$$

$$P^t = \iint_\Omega dP = \iint_\Omega (dP_y^1 + dP_z + dP_y^2) = P_y^{t1} + P_z^t + P_y^{t2}. \tag{10}$$

To calculate (7) and (8),$\langle|Q^t|^2\rangle$,$\left\langle|P^t|^2\right\rangle$, and $\langle P^t Q^{t*}\rangle$have to be calculated first. For example, $\langle|Q^t|^2\rangle$ is given by

$$\begin{aligned}\left\langle|Q^t|^2\right\rangle &= \langle Q^t Q^{t*}\rangle = \langle (Q_y^{t1} + Q_z^t + Q_y^{t2}) \\ &\quad \times (Q_y^{t1} + Q_z^t + Q_y^{t2})^*\rangle \\ &= \langle Q_y^{t1} Q_y^{t1*}\rangle + \langle Q_y^{t1} Q_z^{t*}\rangle + \langle Q_y^{t1} Q_y^{t2*}\rangle + \langle Q_z^t Q_y^{t1*}\rangle \\ &\quad + \langle Q_z^t Q_z^{t*}\rangle + \langle Q_z^t Q_y^{t2*}\rangle + \langle Q_y^{t2} Q_y^{t1*}\rangle \\ &\quad + \langle Q_y^{t2} Q_z^{t*}\rangle + \langle Q_y^{t2} Q_y^{t2*}\rangle . \end{aligned} \tag{11}$$

After a simple algebraic manipulation employing [3, (5)–(13)], we can obtain each term in $\langle|Q^t|^2\rangle$,$\langle|P^t|^2\rangle$, and $\langle P^t Q^{t*}\rangle$. For example, we randomly list one term given as follows:

$$\begin{aligned}\left\langle P_y^{t1} Q_z^{t*}\right\rangle = h\frac{E_0^2}{8\pi}\int_0^{2\pi}\int_0^{\frac{\pi}{2}} \sin^2\alpha\cos\alpha\cos\beta i\sin(kh\cos\alpha) \\ \bullet \frac{\mathrm{e}^{-(\gamma^* - ik\sin\alpha\cos\beta)\mathrm{L}} - 1}{\gamma^* - ik\sin\alpha\cos\beta} d\alpha d\beta.\end{aligned} \tag{12}$$

Obviously, the aforementioned integral is only related to the solid angle. It is very difficult to calculate each term as in (11) analytically. One can take numerical computing environment like MATLAB or make finite summation to get the values of this kind of integrals, which would be time consuming sometimes for a high-level precision.

IV. Some Discussions

A. Relationship Between the Two Semianalytical Solutions

In the previous section, we give a complete derivation of the semianalytical solution which is related to the integrals over the solid angle. In [6], an earlier proposed semianalytical solution was presented but it was related to the geometrical parameters of the TL. A simplification that the horizontal and vertical components of the incident field were independent was made in [6] and several terms were neglected. For example, several terms such as $\left\langle Q_y^1 Q_z^*\right\rangle$,$\left\langle Q_z Q_y^{1*}\right\rangle$,$\left\langle Q_z Q_y^{2*}\right\rangle$, and $\left\langle Q_y^2 Q_z^*\right\rangle$ were neglected in [6, eq. (15)], so the solution of [6] was not complete. In [8], a supplementary spatial correlation function was proposed and can help to amend this simplification; thus, the solution of [6] with the contributions of the supplementary spatial correlation function became complete. It seems that these two semianalytical solutions are different. However, these two solutions come from a same original expression but are derived to different forms. In this sense, both solutions are equivalent to each other.

For simplicity, we just take one term for instance as follows:

$$\begin{aligned}\left\langle Q_y^{t1} Q_y^{t1*}\right\rangle &= \left\langle \left(\iint_{\Omega_1} dQ_{y_1}^1\right) \bullet \left(\iint_{\Omega_2} dQ_{y_2}^1\right)^* \right\rangle \\ &= \left\langle \left(\iint_{\Omega_1} \left(-\frac{1}{2}\int_0^h F_y(0,y_1,0)\exp(i\vec{k}\bullet\vec{r}_1)dy_1\right) d\Omega_1\right) \right. \\ &\quad \left. \bullet \left(\iint_{\Omega_2} \left(-\frac{1}{2}\int_0^h F_y(0,y_2,0)\exp(i\vec{k}\bullet\vec{r}_2)dy_2\right) d\Omega_2\right)^* \right\rangle .\end{aligned} \tag{13}$$

Because the integrals related to the geometrical parameters have nothing to do with the integrals related to the solid angle, the aforementioned expression can also be rewritten as

$$\begin{aligned}&\left\langle Q_y^{t1} Q_y^{t1*}\right\rangle \\ &= \frac{1}{4}\left\langle \left(\iint_{\Omega_1} \left(\int_0^h F_y(0,y_1,0)\exp(i\vec{k}\bullet\vec{r}_1)dy_1\right) d\Omega_1\right) \right. \\ &\quad \left. \bullet \left(\iint_{\Omega_2} \left(\int_0^h F_y^*(0,y_2,0)\exp(-i\vec{k}\bullet\vec{r}_2)dy_2\right) d\Omega_2\right) \right\rangle \\ &= \left\langle \iint_{\Omega_1} F_{\alpha_1}\sin\alpha_1\sin(kh\cos\alpha_1)/(k\cos\alpha_1)d\Omega_1 \right. \\ &\quad \left. \bullet \iint_{\Omega_2} F_{\alpha_2}^*\sin\alpha_2\sin(kh\cos\alpha_2)/(k\cos\alpha_2)d\Omega_2 \right\rangle \\ &= \mathrm{h}^2\frac{E_0^2}{8\pi}\int_0^{2\pi}\int_0^{\frac{\pi}{2}} \sin^3\alpha\frac{\sin^2(kh\cos\alpha)}{(k\cos\alpha)^2}d\alpha d\beta\end{aligned} \tag{14}$$

or

$$\begin{aligned}\left\langle Q_y^{t1} Q_y^{t1*}\right\rangle &= \frac{1}{4}\int_0^h\int_0^h \left[\left\langle \iint_{\Omega_1} F_y(0,y_1,0)\exp(i\vec{k}\bullet\vec{r}_1)d\Omega_1 \right.\right. \\ &\quad \left.\left. \bullet \iint_{\Omega_2} F_y^*(0,y_2,0)\exp(-i\vec{k}\bullet\vec{r}_2)d\Omega_2 \right\rangle\right] dy_1 dy_2 \\ &= \frac{1}{4}\int_0^h\int_0^h \left\langle E_y(0,y_1,0)E_y^*(0,y_2,0)\right\rangle dy_1 dy_2 \\ &= \frac{1}{4}\frac{E_0^2}{3}\int_0^h\int_0^h [\rho_l(|y_1 - y_2|) + \rho_l(|y_1 + y_2|)]dy_1 dy_2\end{aligned} \tag{15}$$

where ρ_l is detailed in [4]. As shown in (14), we can first calculate the integrals that are related to the length and/or the height of the TL analytically as in (6) and then obtain the final expressions consisting of integrals over the solid angle. But in the earlier paper [6], the integrals that are related to the solid angle were calculated in advance; therefore, many useful spatial correlation functions were obtained. With the help of these correlation functions, the solution was only related to the integrals over the length and/or the height of the TL, as shown in (15).

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Owing to the different derivation of the two semianalytical solutions, we can understand the coupling from different physical aspects and obtain the same results.

B. Analytical Solution

For the model detailed in Section II, it is reasonable to apply the approximations $\gamma = jk$ and $\sin(kh\cos\alpha) \approx kh\cos\alpha$ to some terms; thus, we have

$$\left\langle \left|Q^t\right|^2 \right\rangle = \left\langle \left|P^t\right|^2 \right\rangle = h^2 \frac{E_0^2}{8\pi} \int_0^{2\pi} \int_0^{\frac{\pi}{2}} 2\sin\alpha \left\{1 - \cos\left[kL(1+\sin\alpha\cos\beta)\right]\right\} d\alpha d\beta \tag{16}$$

$$\begin{aligned}\left\langle P^t Q^{t*} \right\rangle = h^2 \frac{E_0^2}{8\pi} \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \left[1 - \frac{2\cos^2\alpha}{1-\sin^2\alpha\cos^2\beta}\right] \sin\alpha \\ \bullet\{1 - 2[\cos(kL)\cos(kL\sin\alpha\cos\beta) \\ +2\sin(kL)\sin(kL\sin\alpha\cos\beta)] + \cos(2kL) \\ -i\left[2\sin(kL)\cos(kL\sin\alpha\cos\beta)\right] + i\sin(2kL)\} d\alpha d\beta.\end{aligned} \tag{17}$$

With the aid of the following identities:

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} 2\sin\alpha\cos(kL\sin\alpha\cos\beta) d\alpha d\beta = 4\pi \frac{\sin(kL)}{kL} \tag{18}$$

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} 2\sin\alpha\sin(kL\sin\alpha\cos\beta) d\alpha d\beta = 0 \tag{19}$$

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \sin\alpha \frac{2\cos^2\alpha}{1-\sin^2\alpha\cos^2\beta} d\alpha d\beta = 2\pi \tag{20}$$

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \frac{2\sin\alpha\cos^2\alpha\cos(kL\sin\alpha\cos\beta)}{1-\sin^2\alpha\cos^2\beta} d\alpha d\beta = 2\pi \frac{\sin(kL)}{kL} \tag{21}$$

$$\int_0^{2\pi} \int_0^{\frac{\pi}{2}} \frac{2\sin\alpha\cos^2\alpha\sin(kL\sin\alpha\cos\beta)}{1-\sin^2\alpha\cos^2\beta} d\alpha d\beta = 0 \tag{22}$$

we finally obtain

$$\left\langle \left|Q^t\right|^2 \right\rangle = \left\langle \left|P^t\right|^2 \right\rangle = \frac{h^2 E_0^2}{4} \left[2 - \sin(2kL)/kL\right] \tag{23}$$

$$\left\langle P^t Q^{t*} \right\rangle = 0. \tag{24}$$

It is worthy to note that identity (18) was derived based on the following identities detailed in [10]–[12], respectively, which are given by

$$J_v(z) = \frac{(z/2)^v}{\Gamma(v+1/2)\Gamma(1/2)} \int_0^{\pi} \sin^{2v}\theta\cos(z\cos\theta) d\theta \quad \mathrm{Re}(\nu) > -1/2 \tag{25}$$

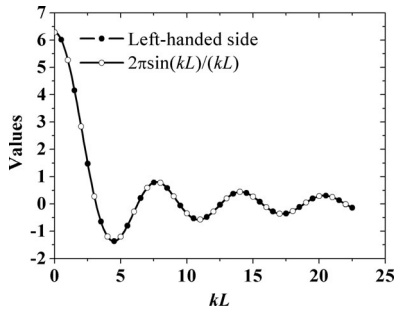


Fig. 2. Values of an integral with a varying argument.

$$\int_0^{\pi} J_0(z\cos\theta)\sin\theta d\theta = \sqrt{\frac{2\pi}{z}} J_{1/2}(z) \tag{26}$$

$$J_{1/2}(z) = \sqrt{\frac{2}{\pi z}} \sin z \tag{27}$$

where J_v is the Bessel function of order ν. Identities (19) and (22) were obtained by analyzing the properties of the integrands over their own interval without any calculation. However, both (20) and (21) seem much more complicated and it is very difficult to calculate them analytically. Therefore, we had to take numerical methods to calculate them, thus got (20) easily, as well as (21) fortunately by observing the numerical values of the integral versus variation of the product of $k$and L. In particular, we have calculated a lot of values of both sides of (21) numerically, as shown in Fig. 2. From Fig. 2, we can see that the two curves agree so well that we can think that identity (21) is accurate although we have not proved it analytically.

C. Considerations on the Solutions

Up to now, there have been three solutions with different forms, the first one in [6] and the last two in this paper, to the problems of field coupling to a single wire over a ground plane in an electrically large cavity. However, these solutions have to face the constrains that:

1) the realistic configurations have to be a uniform TL consisting of an unshielded single conductor near a PEC ground plane as well as two parallel wires far away from the ground plane;
2) all the solutions are based on the classical TL theory [5]; in order to give precise results, the TL cross section has to be electrically small enough at the frequency of interest;
3) the TL is assumed to be free of radiation;
4) for the analytical one, the wire must be lossless, and the wave along the line is assumed propagating with the speed of light;
5) for the single-wire configuration, the plane waves are assumed strictly from the upper half-space, while the whole space for the double-wire configuration.

Owing to the compact forms in (23) and (24), we can solve problems of field coupling to wire more easily and quickly in most cases. From (7), (8), (23), and (24), it is obvious that the coupling is dependent on the height, the electrical length, and the loads of the TL as the following.

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It seems that the coupling is proportional to the square of the height from (23), but it is not the case because actually the characteristic impedance is obviously dependent on the height; thus, reflected coefficients at terminals are also influenced by the height. However, taking matched cases into account, we have

$$\left\langle |I(0)|^2 \right\rangle = \left\langle |I(L)|^2 \right\rangle = \left(E_0^2/4\right)(h/Z_c)^2\left[2-\sin(2kL)/(kL)\right]. \quad (28)$$

From (28), we can see that the lower the height, the weaker the coupling. It is very interesting to compare (28) with [7, eq. (30)]. The former is only available to the single-wire TL, whereas the latter to the double-wire TL.

As for the length, take the matched cases for instance, as shown in (28) or [7, eq. (30)], the coupling increases with the line length and becomes maximal when the length approximates a third of the wavelength, and then reaches a constant plateau for longer lines. For the mismatched cases, how the length affects the coupling will be shown in the numerical examples in the next section.

As for the loads, it has to compare them with the characteristic impedance of the TL. If the load at one end is much smaller or greater than the characteristic impedance, the end can be considered to be short or open. These two extreme cases will induce very strong coupling at the other end for some critical line lengths, which will be discussed as well through examples in the next section.

As presented in [7], the single-wire configuration and the double-wire configuration are identical from the TL point of view. So, the semianalytical or analytical solution in this paper can be directly extended or converted for the double-wire configuration. Generally, the mean square of the magnitude of the current for the single-wire configuration is twice as large as that for the double-wire configuration. In particular, the analytical solutions for the single-wire configuration and the double-wire configuration are not only available in matched cases but also in mismatched cases. For the single-wire configuration, the plane waves have been assumed strictly from the half-space [6], but this constraint can be relaxed. It is usually thought that the plane waves are only from the upper half-space at the surface of the ground plane, while from the whole space at the middle area of the working volume. Although the single wire is near the ground plane, there still exists a distance. Therefore, it is reasonably assumed that the plane waves to the wire are from the whole space at the most or from the half-space at the least. Thus, the coupling quantities associated with the energy as in (7) and (8) for the former are twice as large as that for the latter. So, to predict the coupling, we can take both cases into account.

We have developed a semianalytical solution for general configurations as well as an analytical solution for some special cases. The semianalytical solution can be applied whether the wires are lossless or not, while the analytical solution requires that the wires are lossless. However, most realistic configurations can satisfy the requirements of the analytical solution, so we can directly calculate the closed-form formulas to get the values very quickly.

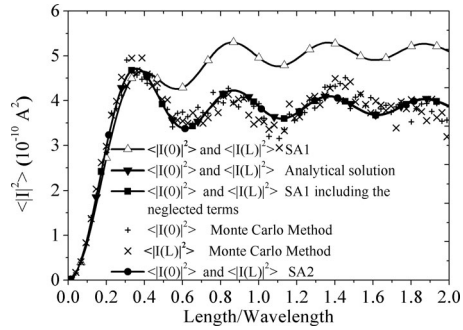


Fig. 3. Comparisons among the solutions.

V. Numerical Examples

In this section, four examples are used to demonstrate the solutions derived in this paper as well as the comparisons with the others.

In the first example, we are interested in the mean squared magnitude of the current, $\langle |I(0)|^2 \rangle$ or $\langle |I(L)|^2 \rangle$, at the end of a 40-cm-long uniform lossless cable, whose diameter is 1 mm, situated at 5 mm above a perfectly conducting ground plane and loaded at both ends by matched loads. On the one hand, $\langle |I(0)|^2 \rangle$ and $\langle |I(L)|^2 \rangle$ are analytically obtained based on (28). On the other hand, $\langle |I(0)|^2 \rangle$ and $\langle |I(L)|^2 \rangle$ are also obtained separately by using the semianalytical method in [6] (designated as SA1), the corrected method in [8], and the semianalytical method proposed in this paper (designated as SA2).

All the aforementioned solutions are compared with the solutions obtained by the Monte Carlo method [13], as shown in Fig. 3. It can be seen from Fig. 3 that when the length is electrically small, all the results are hardly different. It is obvious that the shorter the electrical distance between two positions, the smaller the value of the supplementary spatial correlation function [8]. When the electrical distance between two positions becomes larger, the difference in including the contributions of the supplementary spatial correlation function, and not, is more obvious. Hence, when the wavelength becomes comparable to the length of the TL, the results obtained by the earlier semianalytical solution due to the simplification do not agree well with those obtained by other methods. However, when the neglected terms are included, the results obtained by the earlier semianalytical solution always agree well with the others. In particular, the results from both semianalytical solutions are the same as those from the analytic expression. In addition, from the analytical solution, we can further find that:

1) when the line length approximates a third of the wavelength, the coupling reaches the maximum as long as one end is matched to the characteristic impedance;
2) if one end is matched to the characteristic impedance, then the coupling at the end is strongest when the other end is short or open, and the coupling is the weakest when the other end is matched as well.

In the second example, we calculate the mean of the current amplitude $\langle |I(0)| \rangle$ using $\langle |I(0)| \rangle = \sqrt{\langle |I(0)|^2 \rangle}\sqrt{\pi/4}$ [6], [8] at the end of a 50-cm-long cable, whose diameter is 1 mm,

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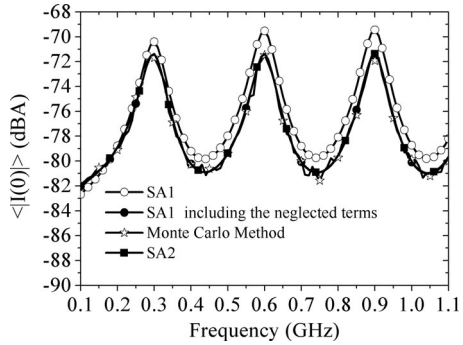


Fig. 4. Means of the current amplitude calculated.

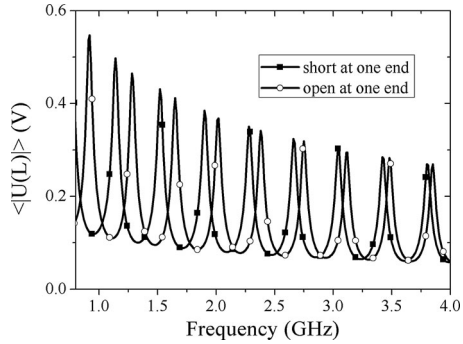


Fig. 5. Mean value of the magnitude of the coupled voltage.

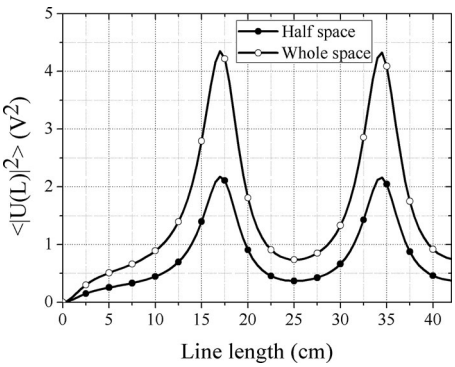


Fig. 6. Mean square value of the magnitude of the coupled voltage.

situated at 3 cm above a perfectly conducting ground plane and loaded at both ends by 50 Ω. For this same problem as shown in [6], the curves in Fig. 4 illustrate that results from the semianalytical solution proposed in this paper agree well with the ones by the Monte Carlo method, as well as ones by the earlier semianalytical solution with the neglected terms included. It is noted that these symmetrical loads are much smaller than the characteristic impedance 405.8 Ω, which can be considered to be short at both ends. For this fixed length, the coupling increases with the increasing frequency, reaches a first maximum when the line length is half of the wavelength, then decreases and goes to a first minimum. It can be summarized that the coupling reaches a maximum at the line length $L = \lambda/2$,λ, $3\lambda/2$, ..., while reaches a minimum at the line length $L = \lambda/4$,$3\lambda/4$,$5\lambda/4$,...

It has to be noted that the constant E_0 is normalized to 1(V/m) in both examples. In the Monte Carlo method in both examples, the convergence was reached by considering 250 successive channel realizations, each realization being simulated by 1000 random plane waves. The plane waves configurations used for all frequencies are the same.

In the third example, we repeat to simulate the cases as shown in [7, Fig. 17]. The TL model is same to that in the first example, but the physical length is $L = 360$ mm, the distance of the wire to the wall is $h = 16$ mm, and the radius of the wire is $r = 0.69$ mm. One end is loaded by $Z_2 = 50\,\Omega$, and the other end is short or open. For this simulation, the power into the chamber is $P_{in} = 12$ dBm, the chamber quality is $Q = 8000$, and the volume of the chamber is $V = 1.62\,\mathrm{m}^3$. So the chamber constant E_0in this example is decided by $E_0^2 = QP_{\mathrm{in}}/(\omega\varepsilon V)$ [7]. As presented in [7], for the short line case, the length used in the simulation is 394 mm; for the open line case, the length used in the simulation is 409 mm. In this example, we can directly take the analytical solution to get the mean value of the magnitude of the voltage $\langle|U(L)|\rangle$ at the end Z_2, and $\langle|U(L)|\rangle = 50\langle|I(L)|\rangle$. As shown in Fig. 5, the frequency ranges from 0.8 GHz to 4 GHz. When this figure is compared with [7, Fig. 17], we can find that the simulated values, in both short and open cases, in Fig. 5 timed by $\sqrt{2}$ may be that in [7, Fig. 17]. This obvious difference is very interesting. However, as presented in Section IV-C, the results of $\langle|I(L)|^2\rangle$ of the single-wire configuration are twice as large as that of the double-wire configuration. But the impedance of the loads of the single-wire configuration is half of that of the double-wire configuration. In [7], it is hard to catch how the double-wire simulations were converted into the single-wire simulation. Maybe the double-wire simulations were directly used. Hence, the factor $\sqrt{2}$ is resulted. However, if we think that the plane waves into the single-wire are from the whole space, the results in Fig. 5 will agree with that in [7, Fig. 17]. Of course, as presented in [7], from Fig. 5, the resonances between the open- and the short-circuited case occur oppositional. The maximum values decrease with the increasing frequency only because that the chamber constant E_0 decreases with the increasing frequency.

It is almost the same as the third example; in the last example, we keep both terminals of the single-wire TL 50 Ω and take the analytical formula to calculate the values of $\langle|U(L)|^2\rangle$ with different lengths at the frequency of 870 MHz under the condition that $P_{\mathrm{in}} = 25$ dBm. The results at both extreme cases are illustrated in Fig. 6. We can see that the results considering that the plane waves coming from the whole space agree well with that simulated in [7, Fig. 16]. Taking the measured results in [7, Fig. 16] into account, we can think that the measured results are between the two curves. Furthermore, let E_0=1 V/m and $Z_2 = 50\Omega$; several curves with different Z_1 are obtained by the analytical solution, as shown in Fig. 7. From Fig. 7, it is demonstrated that a critical line length of $\lambda/4$to $\lambda/2$ actually exists, as presented in [7], depending on the load impedances. At this critical length, the coupling reaches a first maximum. It is also shown that the more the mismatch, the stronger the coupling.

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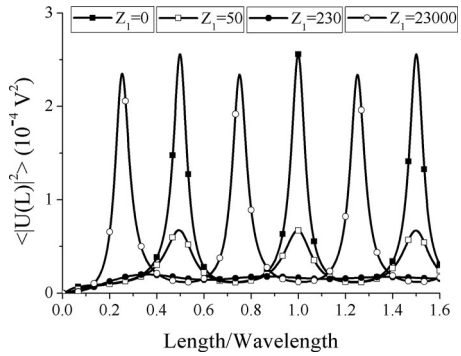


Fig. 7. Mean square value of the magnitude of the coupled voltage with different load impedance at one end.

VI. Conclusion

In this paper, a complete semianalytical solution to problems of field-to-wire coupling in an electrically large cavity is first proposed by using a different method, and the relationship between this solution and an earlier proposed semianalytical solution is discussed as well. When the neglected terms are included in the earlier one, both semianalytical solutions are equivalent to each other, although they are of different forms. Under some reasonable approximations, an analytical solution is obtained straightly from the semianalytical solution which is proposed in this paper. Furthermore, these solutions for the single-wire configuration are also converted to that for the double-wire configuration. Owing to the compact closed-form solution, we can understand the coupling mechanism more easily and clearly.

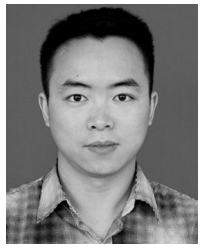
For most realistic configurations like unshielded single-wire or double-wire configuration, the solutions derived in this paper are available. However, more complicated configurations such as multiwire conductors should be further studied.

Acknowledgment

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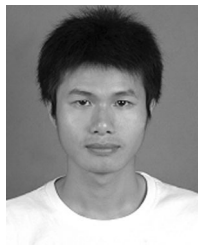
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